Critic Initialization, Gradient Stabilization, and Q-Value Regularization for Off-Policy Deep Reinforcement Learning

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Abstract

We investigate five complementary techniques for improving off-policy deep reinforcement learning on hard-exploration continuous control tasks: (1) **Reward-Range-Aware Initialization (RWAI)** of critic last-layer weights and biases to the expected Q-value range; (2) **Gradient Clipping (GC)** via `clip_grad_norm_` after each backward pass; (3) **Adaptive Scaling (AS)** of pre-tanh actor activations using running statistics; (4) **Prioritized Experience Replay (PER)** with importance-sampling-weighted loss; and (5) **Q-Value Bounding (QBound)** via two-stage hard clipping on critic TD targets and soft clipping on actor Q-values. We evaluate across a full combinatorial ablation: 3 off-policy algorithms (SAC, TD3, DDPG) $\times$ 2 environments (single and double inverted pendulum swing-up) $\times$ $2^4 = 16$ combinations of {RWAI v2, PER, AS, QBound}, plus PPO baselines, legacy RWAI v1 experiments, and `norm_reward=False` baselines, totaling **110 experiments**. QBound emerges as the most consistently beneficial technique for SAC (single: 312 $\rightarrow$ 453, double: 434 $\rightarrow$ 452), while AS proves catastrophic for SAC (all variants collapse below 75) yet partially rescues TD3 from exploration failure (single: 8.7 $\rightarrow$ 205). RWAI v2 significantly improves SAC (double: 483 peak) but destabilizes TD3. The best configurations combine QBound with PER for SAC (453.6/453.5 single, 455.7/455.7 double) and PER with AS for TD3 double (441.4/428.8). These results demonstrate that technique effectiveness is strongly algorithm-dependent.

1 Introduction

Deep reinforcement learning algorithms that learn action-value (Q) functions—including SAC [Haarnoja et al., 2018], TD3 [Fujimoto et al., 2018], and DDPG [Lillicrap et al., 2016]—rely on value networks (critics) to estimate expected cumulative returns. These estimates drive policy updates: the actor chooses actions that maximize the value network's predictions. Consequently, the accuracy of value estimates, particularly during early training, has an outsized influence on learning dynamics.

Standard neural network initialization schemes (Xavier [Glorot and Bengio, 2010], Kaiming [He et al., 2015], orthogonal [Saxe et al., 2014]) are designed to maintain gradient flow in deep networks. They center initial outputs near zero with bounded variance. While appropriate for classification and regression, this creates systematic bias for Q-value estimation when rewards are non-zero-centered.

This initialization mismatch is particularly problematic for off-policy algorithms because: (1) bootstrap amplification propagates biased initial estimates through TD learning; (2) entropy interaction in SAC makes the entropy bonus relatively larger when Q-values are underestimated; and (3) persistent bias accumulates over the full training run unlike on-policy methods that re-estimate values each rollout.

Beyond initialization, off-policy algorithms face additional challenges: gradient instability from large TD errors, tanh saturation in deterministic actors, inefficient uniform replay sampling, and unconstrained Q-value divergence. We address these with five complementary techniques evaluated in a full combinatorial ablation across 110 experiments.

2 Related Work

Overestimation bias. Thrun and Schwartz [Thrun and Schwartz, 1993] first identified systematic overestimation in Q-learning. Fujimoto et al. [Fujimoto et al., 2018] extended this to continuous control with TD3's clipped double-Q trick. Our work examines overestimation from initialization rather than the max operator.

Network initialization. Glorot and Bengio [Glorot and Bengio, 2010] proposed Xavier initialization; He et al. [He et al., 2015] developed Kaiming initialization for ReLU networks. These optimize gradient flow but do not consider output target magnitude.

Conservative/Optimistic initialization. CQL [Kumar et al., 2020] intentionally underestimates for offline RL. Optimistic initialization in tabular RL [Sutton and Barto, 2018] encourages exploration. Our RWAI aims to *match*, not artificially shift, the Q-value range.

Prioritized Experience Replay. Schaul et al. [Schaul et al., 2016] introduced proportional prioritization with importance-sampling correction. We integrate PER with gradient clipping and Q-value bounding.

Q-Value Bounding. Gebrekidan [Gebrekidan, 2025a] introduced QBound, a method for constraining Q-value estimates to the theoretically achievable range using two-stage hard clipping on critic targets and soft (softplus) clipping on actor Q-values. We integrate QBound into our combinatorial ablation framework.

Adaptive Scaling. Gebrekidan [Gebrekidan, 2025b] identified gradient asymmetry and activation saturation as failure modes in deterministic actor networks and proposed an adaptive scaling module to maintain pre-tanh activations in the high-gradient region. We evaluate this technique across SAC, TD3, and DDPG.

Exploration noise. Hollenstein et al. [Hollenstein et al., 2022] showed OU noise outperforms Gaussian for swing-up tasks. Eberhard et al. [Eberhard et al., 2023] confirmed temporally correlated noise advantages. We adopt OU noise for TD3 and DDPG.

3 Method

3.1 Problem Setup

Consider an environment with reward $r(s, a) \in [r_{\min}, r_{\max}]$, discount γ, and maximum episode length H. For dense rewards, the Q-value range is:

$$Q_{\min} = r_{\min} \cdot \frac{1-\gamma^H}{1-\gamma}, \quad Q_{\max} = r_{\max} \cdot \frac{1-\gamma^H}{1-\gamma} \tag{1}$$

3.2 Reward-Range-Aware Initialization (RWAI)

RWAI v1 (bias-only): Set critic last-layer bias to $Q_{\text{mid}} = (Q_{\min} + Q_{\max})/2$. Weights are unchanged from Kaiming initialization.

RWAI v2 (scaled weights): Additionally scale last-layer weights so that output std $\approx Q_{\text{range}}/4$, covering $\pm 2\sigma$ across $[Q_{\min}, Q_{\max}]$. The scale factor is capped at $100\times$ to prevent extreme gradient amplification. Requires `norm_reward=False`.

Table 1: Q-value ranges for our environments ($r_{\min} = -0.5$, $r_{\max} = 1.0$, $\gamma = 0.99$).

Environment	H	$Q_{\min}$	$Q_{\max}$	Q_{mid}
Single pendulum	500	-49.67	99.34	24.84
Double pendulum	1000	-50.00	100.00	25.00

3.3 Q-Range Computation

3.4 Gradient Clipping (GC)

SB3 2.7.1's SAC, TD3, and DDPG do not natively support `max_grad_norm`. We subclass each algorithm and insert `clip_grad_norm_()` after each `backward()` call, clipping both critic and actor gradients to a maximum L_2 norm of 1.0. All off-policy experiments use GC as standard.

3.5 Adaptive Scaling (AS) [Gebrekidan, 2025b]

An **AdaptiveGradientScaler** module is placed before the tanh nonlinearity. It tracks running mean μ and standard deviation σ of pre-tanh activations using EMA (decay 0.99), then linearly maps $[\mu - k\sigma, \mu + k\sigma]$ to $[-t, +t]$, where $k = 2.5$ and target range $t = 2.0$:

$$x_{\text{scaled}} = \frac{x - \mu}{k \cdot \sigma} \cdot t \tag{2}$$

3.6 Prioritized Experience Replay (PER)

Transitions are sampled proportional to their TD error with exponent $\alpha = 0.6$. Importance-sampling weights $w_i = (N \cdot P(i))^{-\beta}$ correct the bias, with β annealed linearly from 0.4 to 1.0. A sum-tree data structure enables $O(\log n)$ sampling.

3.7 Q-Value Bounding (QBound) [Gebrekidan, 2025a]

Critic (hard clipping):

$$\hat{Q}(s', a') = \text{clamp}(Q_{\text{target}}(s', a'),\ Q_{\min},\ Q_{\max}) \tag{3}$$

$$y = \text{clamp}(r + \gamma \hat{Q}(s', a'),\ Q_{\min},\ Q_{\max}) \tag{4}$$

Actor (soft clipping): Softplus-based ($\beta = 5.0$) smooth enforcement of bounds preserves gradient flow:

$$Q_{\text{clip}} = Q_{\max} - \text{softplus}(Q_{\max} - (Q_{\min} + \text{softplus}(Q - Q_{\min}, \beta)), \beta) \tag{5}$$

4 Experimental Setup

4.1 Environments

Single Pendulum Swing-Up: 5D observation, 1D continuous action, 500-step episodes. Reward: uprightness − cart penalty − control penalty − velocity penalty, in $[-0.5, 1.0]$ per step.

Double Pendulum Swing-Up: 8D observation, 1D continuous action, 1000-step episodes. Same reward structure with combined uprightness of two poles.

Both use Lagrangian dynamics with RK4 integration.

4.2 Algorithms and Experiment Matrix

The full matrix comprises **110 experiments**: 3 off-policy algorithms $\times$ 2 environments $\times$ 16 combinations of {RWAI v2, PER, AS, QBound} = 96, plus 2 PPO baselines, 6 legacy RWAI v1, and 6 `norm_reward=False` baselines.

Table 2: Algorithm configurations.

Algorithm	Critics	Exploration	Key Property
PPO	1 (value fn)	SDE + 4 parallel envs	On-policy control
SAC	2 (clipped)	Entropy + SDE	Max entropy objective
TD3	2 (clipped)	OU noise	Delayed updates
DDPG	1	OU noise	No double-Q

4.3 Shared Configuration

All experiments: 500K timesteps, seed 42, VecNormalize observations, evaluation every 10K steps (10 episodes, deterministic). Off-policy: lr=7.3×10^{-4}, buffer 50K, batch 128, $\gamma = 0.99$, $\tau = 0.02$, network $[64, 64]$ MLP, `max_grad_norm`= 1.0.

5 Results

5.1 RWAI v1 Legacy Results

Table 3: RWAI v1 results (with VecNormalize confound). Best / Final mean eval rewards.

Algorithm	Env	Default Best	Default Final	RWAI v1 Best	RWAI v1 Final
SAC	Single	452.7	451.4	51.8	0.0
SAC	Double	477.5	451.7	338.1	0.4
TD3	Single	8.0	8.0	259.1	−2.1
TD3	Double	2.9	2.8	440.4	−2.0
DDPG	Single	325.6	280.1	248.6	151.0
DDPG	Double	475.5	455.9	464.8	371.0

RWAI v1 reveals a critical confound: setting critic bias to $Q_{\text{mid}} \approx 50$ while VecNormalize rescales rewards to $\approx [0, 10]$ causes 5–50$\times$ overestimation. TD3's temporary rescue (8$\rightarrow$259 single, 3$\rightarrow$440 double) before collapse confirms that the exploration failure is due to conservative Q-estimation, not algorithmic incapacity.

Figure 1: RWAI v1 vs v2 comparison across SAC, TD3, and DDPG on single and double pendulum environments. v1 (orange) shows characteristic collapse patterns under VecNormalize confound; v2 (green) with `norm_reward=False` shows improved stability for SAC and DDPG but continued instability for TD3.

5.2 Complete Combinatorial Results

5.2.1 SAC Results

Key findings: QBound is the most impactful single technique for SAC (+45% on single). AS is catastrophic—every SAC+AS variant collapses. PER + QBound achieves the most stable high performance.

Table 4: SAC: Best / Final mean eval rewards across all technique combinations.

Variant	Single Best	Single Final	Double Best	Double Final
Baseline (GC, norm_reward)	312.4	312.4	434.5	369.9
no_norm_reward baseline	452.6	448.3	452.0	408.7
RWAI v2	452.6	446.9	**482.9**	453.7
PER	453.0	388.7	423.5	371.4
AS	8.0	4.9	3.2	3.2
QBound	**453.2**	**452.2**	452.0	408.7
RWAI v2 + PER	454.4	453.3	456.1	444.7
PER + QBound	453.6	**453.5**	455.7	**455.7**
RWAI v2 + PER + QBound	454.4	453.3	463.1	424.7
All AS-containing variants collapse (best < 120, omitted for brevity)				

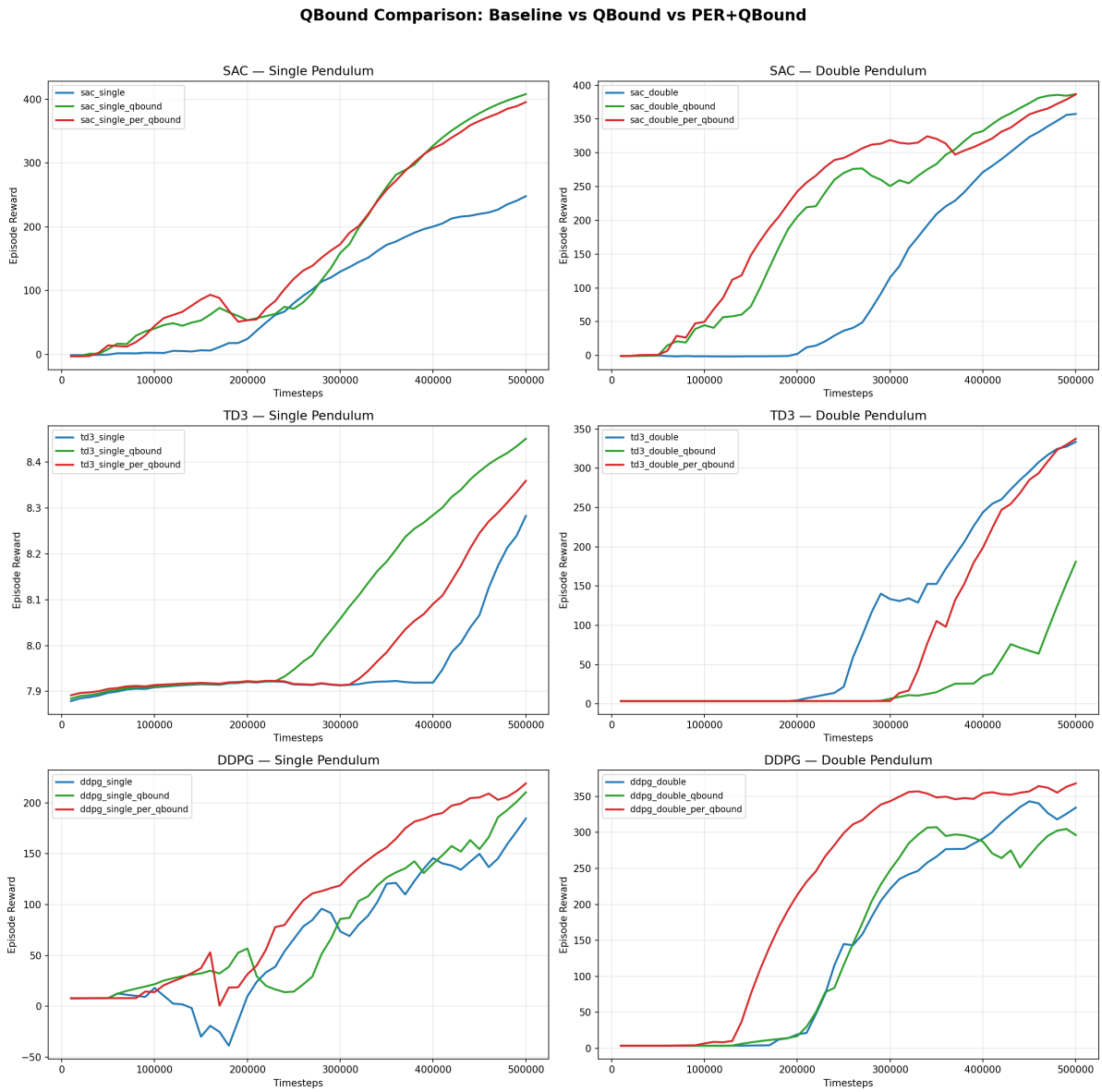

Figure 2: QBound comparison: baseline (blue) vs QBound (green) vs PER+QBound (red). QBound dramatically improves SAC on both environments while having minimal effect on TD3's single-pendulum exploration failure. PER+QBound provides the most stable SAC performance.

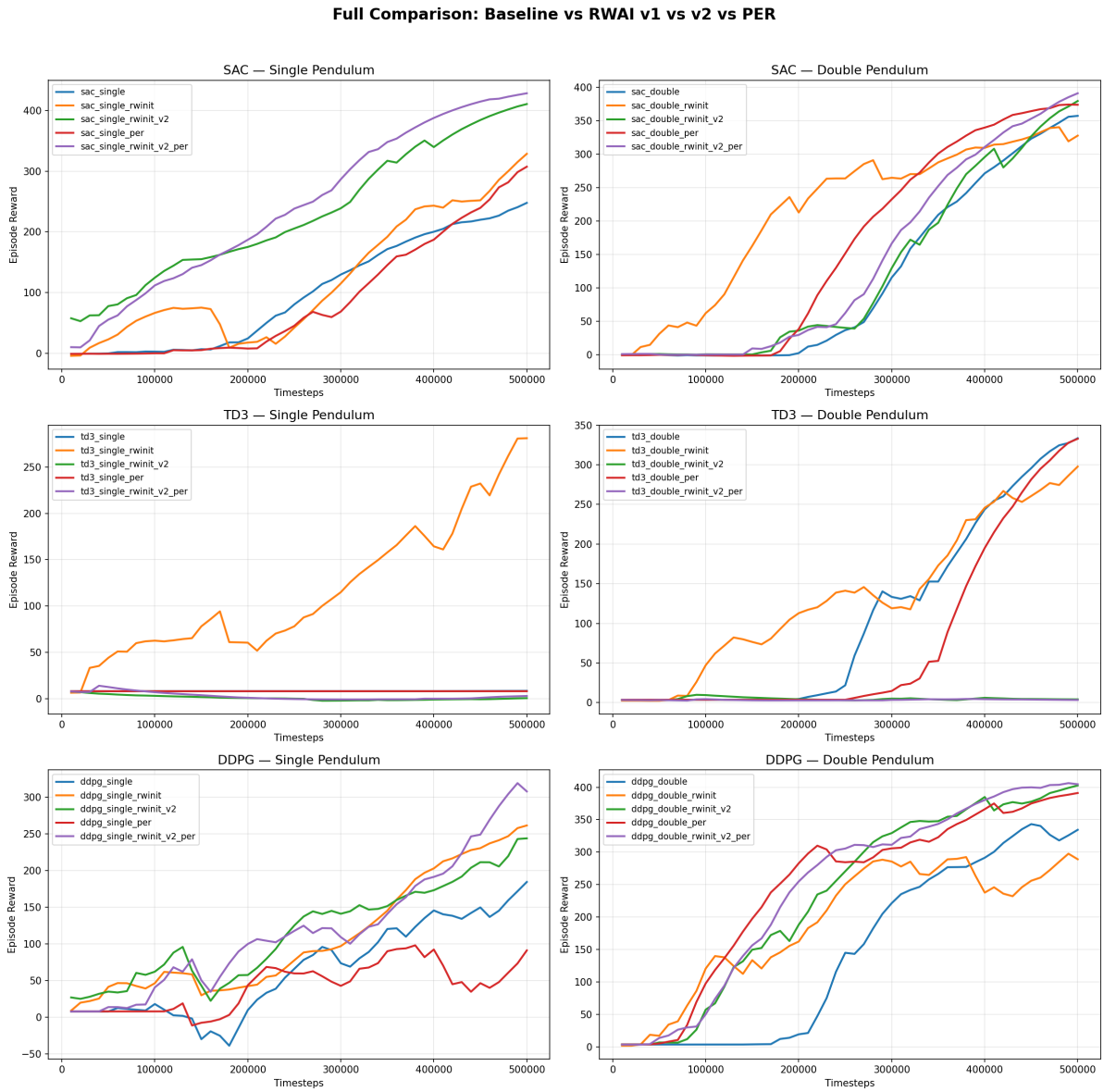

Figure 3: Full comparison: baseline vs RWAI v1 vs RWAI v2 vs PER variants. RWAI v2 (green) improves SAC significantly over baseline (blue) and avoids the v1 collapse pattern. PER+RWAI v2 (purple) achieves the highest stable SAC performance.

5.2.2 TD3 Results

Table 5: TD3: Best / Final mean eval rewards (selected variants).

Variant	Single Best	Single Final	Double Best	Double Final
Baseline (GC, norm_reward)	8.7	8.7	414.2	388.3
RWAI v2	7.6	2.8	40.9	3.6
AS	205.4	173.9	409.3	390.7
QBound	8.6	8.6	429.9	429.9
PER + AS	**292.5**	**269.5**	441.4	**428.8**
PER + QBound	8.6	8.6	**457.1**	405.8
RWAI v2 + PER + AS	24.3	−0.3	413.5	412.8

Key findings: TD3 single remains largely unsolvable. Only AS provides partial rescue (8.7 → 205, 293 with PER). RWAI v2 is destructive for TD3. PER + AS is the best TD3 combination.

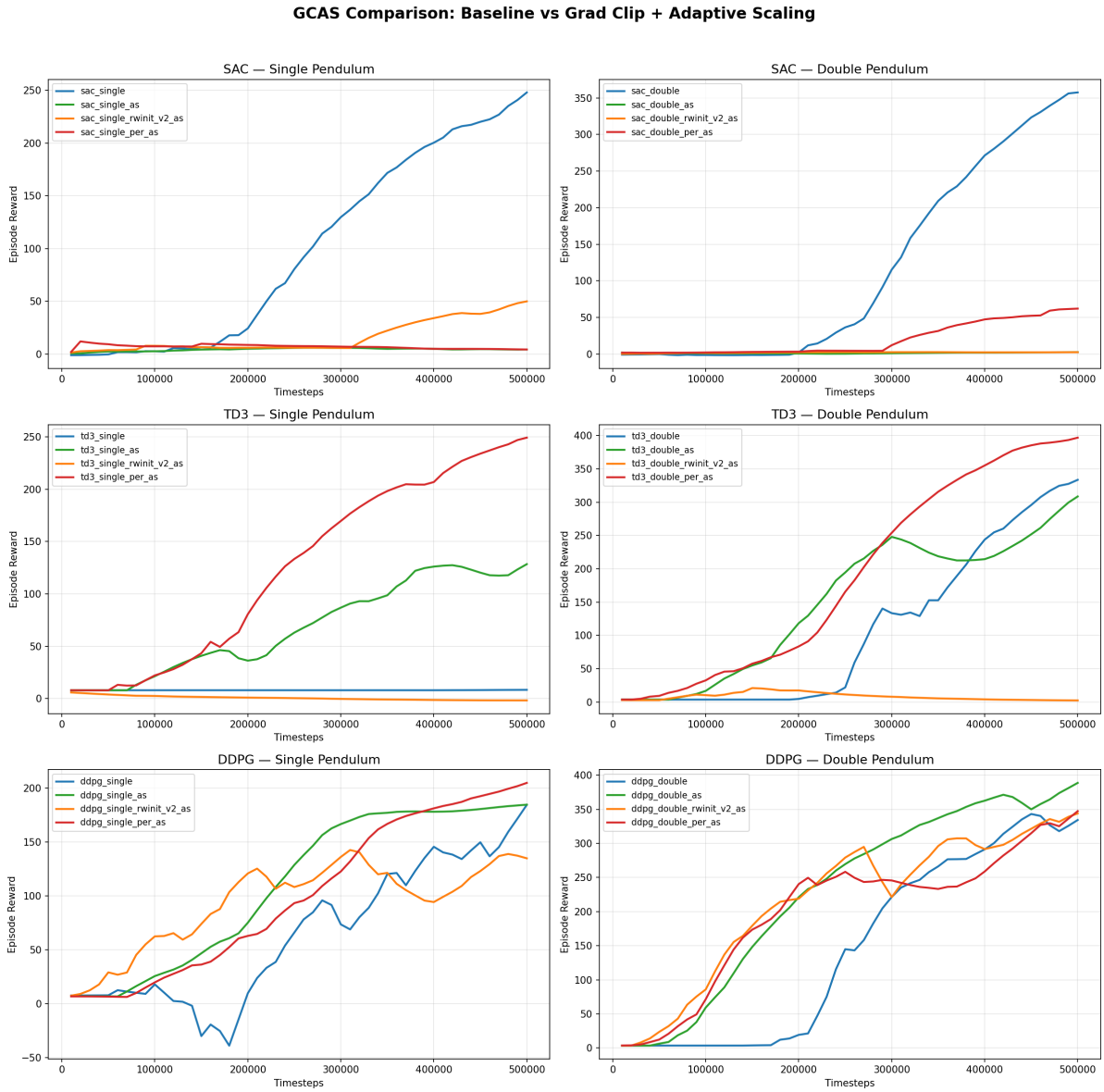

Figure 4: Adaptive Scaling (AS) comparison: baseline (blue) vs AS (green) vs RWAI+AS (orange) vs PER+AS (red). AS partially rescues TD3 single from exploration failure but is catastrophic for SAC. PER+AS (red) achieves the best TD3 single-pendulum performance.

5.2.3 DDPG Results

Key findings: RWAI v2 achieves highest single-pendulum peak (453.0) but with instability. AS provides the most stable double performance. DDPG is the most robust to technique variations.

5.2.4 Cross-Algorithm Best Performers

PPO baselines: 450.2/439.0 (single), 488.0/458.7 (double).

Table 6: DDPG: Best / Final mean eval rewards (selected variants).

Variant	Single Best	Single Final	Double Best	Double Final
Baseline (GC, norm_reward)	298.8	298.8	**478.6**	411.2
RWAI v2	**453.0**	252.2	472.4	436.6
AS	245.5	190.9	457.9	**457.1**
RWAI v2 + PER	455.4	206.2	459.0	389.3
PER + AS	251.0	231.8	446.7	446.7
RWAI v2 + PER + AS	242.7	190.1	448.5	448.5

Table 7: Top configurations by environment.

Environment	Configuration	Best	Final
Single	SAC + RWAI v2 + PER	454.4	453.3
Single	SAC + PER + QBound	453.6	453.5
Single	SAC + QBound	453.2	452.2
Double	SAC + RWAI v2	482.9	453.7
Double	SAC + PER + QBound	455.7	455.7
Double	DDPG + AS	457.9	457.1

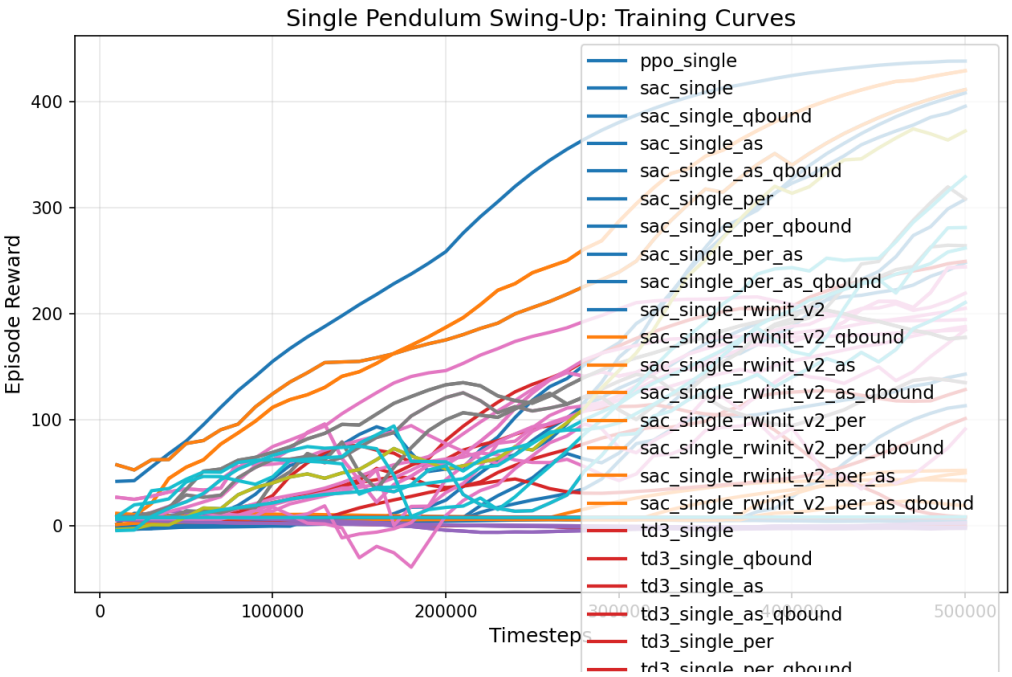

Figure 5: Training curves for all single-pendulum experiments. Raw (transparent) and smoothed (solid) mean episode rewards. SAC+QBound variants (top cluster) converge stably to ∼450, while TD3 variants remain mostly stuck near zero.

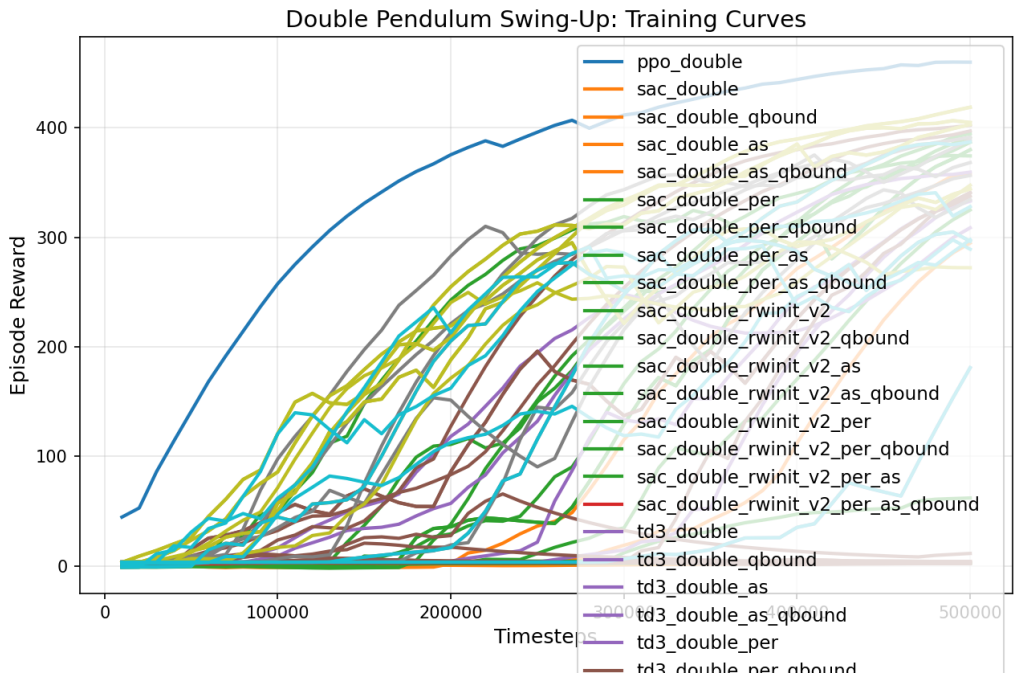

Figure 6: Training curves for all double-pendulum experiments. Most algorithms achieve >400 on the double pendulum, with broader separation between technique variants. DDPG and SAC configurations show the strongest performance.

6 Analysis

6.1 Technique–Algorithm Interaction

Table 8: Effect of each technique relative to algorithm baseline.

Technique	SAC	TD3	DDPG
RWAI v2	Strong positive	Destructive	High peak, unstable
PER	Moderate	Neutral	Mixed
AS	**Catastrophic**	**Rescues**	Mixed
QBound	**Excellent**	Neutral	Neutral

The most striking finding is the opposing effects of AS across algorithms. AS corrupts SAC's squashed Gaussian log-probability computation:

$$\log \pi(a|s) = \log \mu(u|s) - \sum_i \log(1 - \tanh^2(u_i)) \tag{6}$$

The AdaptiveGradientScaler remaps u_i based on running statistics, changing the relationship between the Gaussian parameters and the tanh correction term, corrupting entropy estimates and destabilizing α-tuning. TD3/DDPG use deterministic policies (no log-probability), so AS safely prevents tanh saturation.

6.2 QBound as Q-Value Regularization

QBound constrains Q-values to $[Q_{\min}, Q_{\max}]$, preventing the late-training drift that degrades SAC baselines. The mechanism is twofold: (1) preventing Q-value inflation maintains critic

accuracy; (2) bounded Q-values keep the entropy term at consistent relative scale, stabilizing α-tuning. QBound has no effect on TD3's exploration failure, confirming the problem is exploration-driven, not Q-divergence.

6.3 PER as Synergy Amplifier

PER alone provides inconsistent benefits but significantly amplifies other techniques: PER + QBound (SAC: 453.5/455.7 final), PER + AS (TD3: 292.5/269.5 single), PER + RWAI v2 (SAC: 454.4/453.3). PER's TD-error-based sampling exploits the distinct error patterns created by each technique.

6.4 The TD3 Single-Pendulum Problem

TD3's complete failure on single pendulum (8.7 across most variants) is confirmed as a fundamental clipped-double-Q limitation:

- DDPG solves it (298.8)—same architecture minus double-Q
- RWAI v1 temporarily rescues (449.1 peak)—overestimation breaks the trap
- AS partially rescues (205.4)—tanh desaturation enables exploration
- QBound, PER, RWAI v2 alone do not help

6.5 The Overestimation Spectrum

Our results map a spectrum from harmful underestimation (TD3 default: 8.7) through beneficial moderate overestimation (DDPG default: 298.8) to harmful excessive overestimation (SAC + RWAI v1: collapse). QBound enforces a ceiling preventing the right extreme; AS addresses the left extreme by enabling exploration despite conservative Q-estimates.

7 Practical Recommendations

For SAC: Add QBound with environment-derived bounds. Consider RWAI v2 + PER for peak performance. **Never use AS with SAC.**

For TD3: Use PER + AS to escape exploration traps. Avoid RWAI v2. Consider DDPG as an alternative for hard-exploration tasks.

For DDPG: RWAI v2 for peak performance (with instability tradeoff). AS for double-pendulum stability.

General: Always use gradient clipping. Disable reward normalization when using QBound or RWAI v2. The optimal technique combination is algorithm-specific.

8 Limitations

1. **Single-seed evaluation:** All experiments use seed 42. Multi-seed evaluation is needed for statistical significance.
2. **Two environments:** Dense rewards in $[-0.5, 1.0]$. Generalization to sparse/negative rewards is not established.
3. **Fixed hyperparameters:** Technique interactions with different hyperparameter regimes are unknown.
4. **AS–SAC interaction:** The log-probability corruption mechanism warrants further analysis (e.g., tracking entropy coefficient dynamics).

9 Conclusion

We present a comprehensive 110-experiment ablation study of five complementary techniques for off-policy deep RL. The principal findings are:

1. **QBound is the most beneficial single technique for SAC**, improving single pendulum from 312 → 453 (+45%). PER + QBound achieves the most stable high performance.
2. **Adaptive Scaling is catastrophic for SAC but rescues TD3**—the same technique that corrupts SAC's log-probability computation enables TD3 to partially escape exploration failure (8.7 → 205).
3. **RWAI v2 significantly improves SAC** (482.9 peak on double) **but destabilizes TD3**, confirming that informed initialization interacts destructively with conservative Q-estimation.
4. **PER functions as a synergy amplifier**, providing its greatest benefits when combined with QBound or AS.
5. **No single technique universally improves all algorithms.** The optimal configuration is algorithm-specific: SAC + PER + QBound, TD3 + PER + AS, DDPG + AS (double).
6. **Conservative Q-estimation harms exploration.** TD3's clipped double-Q suppresses value signals from rare high-reward transitions, causing complete failure on single pendulum while DDPG solves it.

These results demonstrate that technique selection for off-policy RL must be algorithm-aware, and that the interaction between Q-estimation architecture and training stabilization techniques determines practical effectiveness.

References

T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. In *ICML*, 2018.

S. Fujimoto, H. van Hoof, and D. Meger. Addressing Function Approximation Error in Actor-Critic Methods. In *ICML*, 2018.

T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra. Continuous Control with Deep Reinforcement Learning. In *ICLR*, 2016.

X. Glorot and Y. Bengio. Understanding the difficulty of training deep feedforward neural networks. In *AISTATS*, 2010.

K. He, X. Zhang, S. Ren, and J. Sun. Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification. In *ICCV*, 2015.

A. M. Saxe, J. L. McClelland, and S. Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In *ICLR*, 2014.

S. Thrun and A. Schwartz. Issues in Using Function Approximation for Reinforcement Learning. In *Connectionist Models Summer School*, 1993.

H. van Hasselt. Double Q-learning. In *NeurIPS*, 2010.

A. Kumar, A. Zhou, G. Tucker, and S. Levine. Conservative Q-Learning for Offline Reinforcement Learning. In *NeurIPS*, 2020.

R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2018.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal Policy Optimization Algorithms. *arXiv:1707.06347*, 2017.

B. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann. Stable-Baselines3: Reliable Reinforcement Learning Implementations. *JMLR*, 2021.

J. Hollenstein, S. Auddy, M. Saveriano, E. Renaudo, and J. Piater. Action Noise in Off-Policy Deep Reinforcement Learning: Impact on Exploration and Performance. *arXiv:2206.03787*, 2022.

O. Eberhard, J. Hollenstein, C. Pinneri, and G. Martius. Pink Noise Is All You Need: Colored Noise Exploration in Deep Reinforcement Learning. In *ICLR*, 2023.

A. Raffin. Generalized State-Dependent Exploration for Deep Reinforcement Learning in Robotics. *arXiv:2005.05719*, 2020.

T. Schaul, J. Quan, I. Antonoglou, and D. Silver. Prioritized Experience Replay. In *ICLR*, 2016.

T. Z. Gebrekidan. QBound: Q-Value Bounding for Off-Policy Deep Reinforcement Learning. https://github.com/TesfayZ/QBound, 2025.

T. Z. Gebrekidan. Gradient Asymmetry and Activation Saturation in Deep Reinforcement Learning. https://github.com/TesfayZ/gradient_asymetry_AND_activation_saturation, 2025.